

MARCELO VENTURINI

Biomedical Engineer

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PROFILE

Recent M.Sc. Biomedical Engineering graduate with a strong electronics-oriented background and hands-on experience in biomedical instrumentation, electronic measurements, embedded systems, sensor interfacing, and signal processing. Proficient with oscilloscopes, multimeters, signal generators, MATLAB/Simulink, Python, and microcontroller programming.

EDUCATION

M.Sc. in Biomedical Engineering, Politecnico di Torino, Italy 2024 – 2026

Final grade: **109/110**. Thesis: *EMG Signal Deconvolution as Preprocessing for Enhanced Hand Gesture Recognition*.

B.Sc. in Biomedical Engineering, Universidad Nacional de Córdoba, Argentina 2020 – 2024

Final grade: **8.54/10**. Awarded a competitive scholarship for international study at Politecnico di Torino.

Relevant coursework: Digital Electronics, Biomedical Instrumentation, Signal Processing, Embedded Systems.

TECHNICAL SKILLS

Test & Lab	Oscilloscopes, multimeters, signal generators, soldering, electronic measurements, hardware debugging
Programming	MATLAB, Simulink, Python, Assembler, Arduino IDE, Excel
Embedded & HW	ESP32, PIC microcontrollers, sensor interfacing, Multisim
Languages	Spanish (native), English (C1), Italian (fluent)

RELEVANT EXPERIENCE

Teaching Assistant – Digital Electronics 2024

Universidad Nacional de Córdoba

- Supported undergraduate laboratory sessions and practical exercises in digital electronics.
- Assisted students with logic circuits, measurements, debugging, and interpretation of experimental results.

SELECTED PROJECTS

Levodopa Response Assessment Device *Incubated Project*

- Designed a complete analog circuit for sEMG signal acquisition and conditioning.
- Programmed an ESP32 microcontroller using the Arduino IDE for real-time data acquisition.
- Evaluated Parkinson's treatment response through quantitative physiological measures.

Ultrasonic Sensor Interfacing with PIC Microcontroller *Academic Project*

- Programmed a PIC microcontroller in assembler for timing-sensitive ultrasonic acquisition.
- Implemented direct hardware interfacing and low-level embedded debugging.

IoT Health & Home Monitoring Platform *Academic Project*

- Integrated HR, SpO₂, and environmental sensors via MQTT using the SenML standard.
- Architected a RESTful CherryPy microservice backend to manage users, devices, and data flow.
- Designed Telegram bots with ThingSpeak integration for real-time monitoring and reporting.

ADDITIONAL EXPERIENCE

Sailing Coach 2022–2026

Hampton YC (USA), Cold Spring Harbor (USA), Huguenot YC (USA), and Orza Minore (Italy)

Further information on projects, technical documentation and certifications is available in my online portfolio:

<https://marcelo-venturini.github.io/>.

